

Finding Eight-Letter Names in a Tree

Instead of printing out the data in each node as we traverse them, we can use a *depth-first search* to find specific data in a tree data structure. We'll write an algorithm that searches the tree in [Figure 4-4](#) for names that are exactly eight letters long. This is a rather contrived example, but it shows how an algorithm can use tree traversal to retrieve data out of a tree data structure.

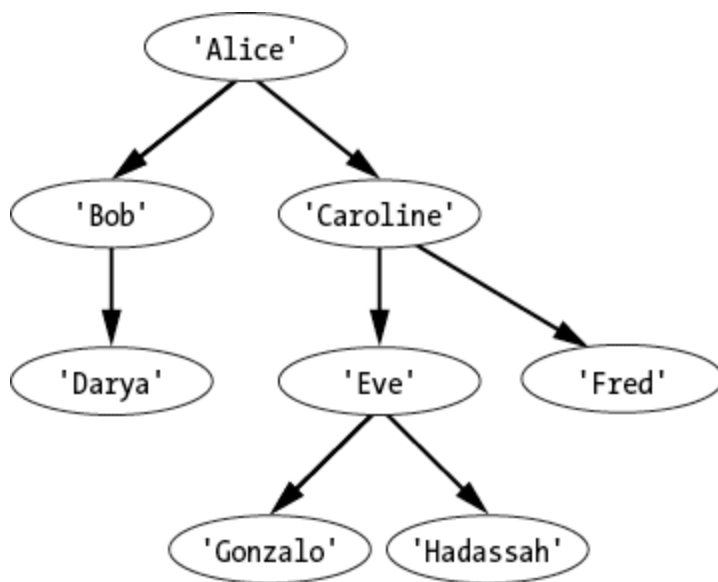


Figure 4-4: The tree that stores names in our `depthFirstSearch.py` and `depthFirstSearch.html` programs

Let's ask the three questions about recursive algorithms for our tree traversal code. Their answers are similar to the answers for the tree traversal algorithms:

What is the base case? Either a leaf node causing the algorithm to backtrack, or a node containing an eight-letter name.

What argument is passed to the recursive function call? The node to traverse to, whose child nodes will be the next nodes to traverse.

How does this argument become closer to the base case? There are no cycles in a DAG, so following the descendant nodes will always eventually reach a leaf node.